Data Science

Clustering

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Outline

1. Introduction

2. Hierarchical clustering
   - Partition, hierarchy and dendrograms
   - Algorithms
   - Partition quality

3. K-means and related methods

Unsupervised learning

Setting

- $\mathcal{D} = ((\mathbf{X}_i)_{1 \leq i \leq N})$
- no target value!
- goal: "understanding" the data
- in practice, many concrete goals such as
  - finding clusters
  - finding frequent patterns
  - finding outliers
  - modeling the data distribution
  - etc.

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Definition

In a data set, a *cluster* is a group of objects that are more similar to each other than to objects from the rest of the data set.

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Difficulties

- group?
- similar?
- more than?

Applications

Simplification (pre-processing)

- large scale data analysis: replace a group of objects by a single typical object
- coarse grain analysis before a finer grain one (see e.g. Cluster17)

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Knowledge discovery

- consumer analysis
- image analysis (zone extraction)
- evolutionary biology
- etc.

How to build a clustering method?

Ingredients

- similarity or dissimilarity: how to compare objects?
- group structure: set, "fuzzy" set, probabilistic membership?
- clustering structure: disjoint sets, overlapping sets, complete partition?

Classical example: standard k-means

- dissimilarity: euclidean distance on numerical variables (i.e. $\mathcal{X} = \mathbb{R}^p$)
- groups: standard sets
- clustering structure: groups for a partition of the data set

Clustering is ill posed

Problems

- vague definition with possible extensions (e.g. find groups that reflect the "underlying structure" of the data set)
- vastly different practical goals (from pre-processing to knowledge discovery)
- no universal quality criterion:
  - almost one criterion per method!
  - lack of task oriented criterion
- impossibility result in some situations

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Clustering as a process

- for knowledge discovery
- usefulness rather than quality

Results and tools

Using a clustering

- cluster analysis: what do they contain?
  - list of the members
  - representative element(s)
- cluster "positioning":
  - relative positioning in space
  - significant differences

Tasks oriented analysis

- can the analyst understand the clusters?
- can the clustering summarize the data? In what sense?

Families of clustering algorithms

Four main families...

- hierarchical algorithms: **hierarchical clustering**
- centroid based algorithms: **k-means**
- density based algorithms (not studied here): DBSCAN
- probabilistic algorithms (not studied here): mixture models with Expectation Maximization algorithm

Families of clustering algorithms

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but a complex landscape

- k-means as a limit case of some mixture models
- DBSCAN is somewhat related to single linkage hierarchical clustering
- numerous hybrid techniques
- etc.

Clustering and dissimilarity

Minimal assumption

- $\mathcal{X}$ is equipped with a dissimilarity d
- d is a dissimilarity on $\mathcal{X}$ iff:
  1. d is a function from $\mathcal{X} \times \mathcal{X}$ to $\mathbb{R}^+$
  2. $\forall \mathbf{X}, \mathbf{X}'$, $d(\mathbf{X}, \mathbf{X}') = d(\mathbf{X}', \mathbf{X})$
  3. $\forall \mathbf{X}, \mathbf{X}'$, $\mathbf{X} \neq \mathbf{X}' \Leftrightarrow d(\mathbf{X}, \mathbf{X}') > 0$

Links with clustering

- many algorithms can work with arbitrary dissimilarities
- density, separability, compactness, in general quality metrics are expressed in terms of the dissimilarity
- results **strongly** depend on the dissimilarity
- plain $\mathbb{R}^R$: euclidean dissimilarity!

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Partitions

Clustering and partition

$\mathcal{D} = ((\mathbf{X}_i)_{1 \leq i \leq N})$

- group: a subset of $\{1, \ldots, N\}$
- clustering structure: each observation (i.e. index in $\{1, \ldots, N\}$) is in one unique group + no empty group
- building a clustering $\Leftrightarrow$ choosing a partition!
- *here* a *cluster* is also a *class*

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Remarks

- this is only *one* possible structure among others!
- one can have $\mathbf{X}_i = \mathbf{X}_j$ for $i \neq j$
  - we expect i and j to be in the same group!
  - groups are described at the index level

Hierarchy

Partial order on partitions

- a partition P is finer than a partition Q ($P \leq Q$) if any class of P is a subset of a class of Q
- example

$$P = \{\{1,2\},\{3,4\},\{5,6\}\} \leq \{\{1,2,3,4\},\{5,6\}\} = Q$$

Hierarchy

A hierarchy for $\mathcal{D} = ((\mathbf{X}_i)_{1 \leq i \leq N})$ is a fully ordered set of partitions containing the trivial partitions

1. $\{\{1,\ldots,N\}\}$
2. $\{\{1\},\{2\},\ldots,\{N\}\}$

Dendrogram

Graphical representation of a hierarchy by a tree

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A B C D E F

Dendrogram

Graphical representation of a hierarchy by a tree

- most refined partition (trivial)

Dendrogram

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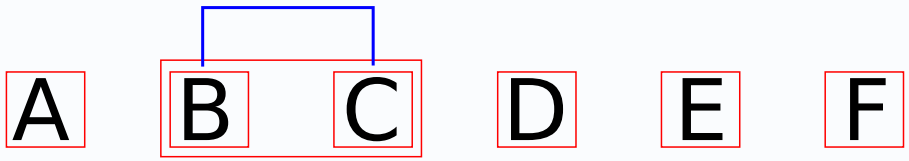

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- B and C are in the same class at level 2: leaves of a common node

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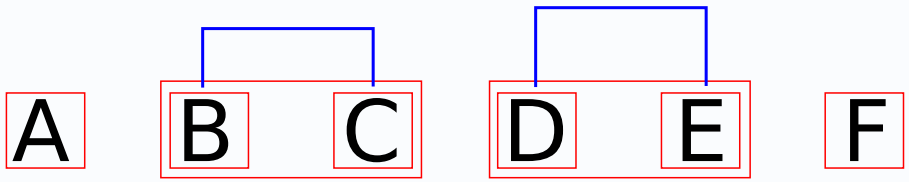

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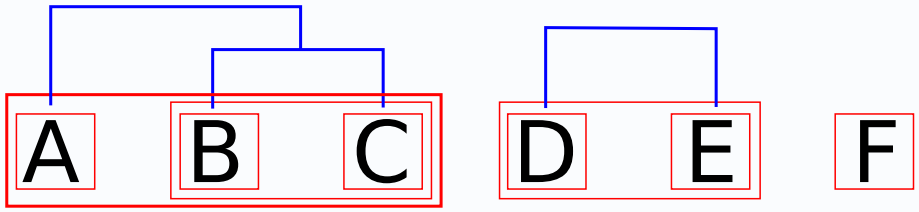

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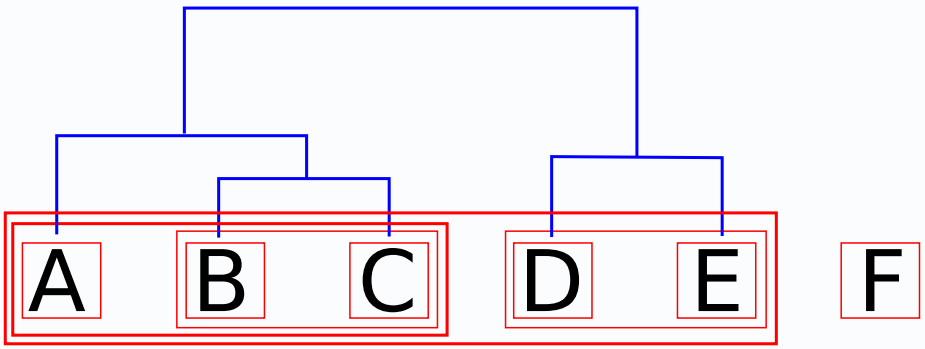

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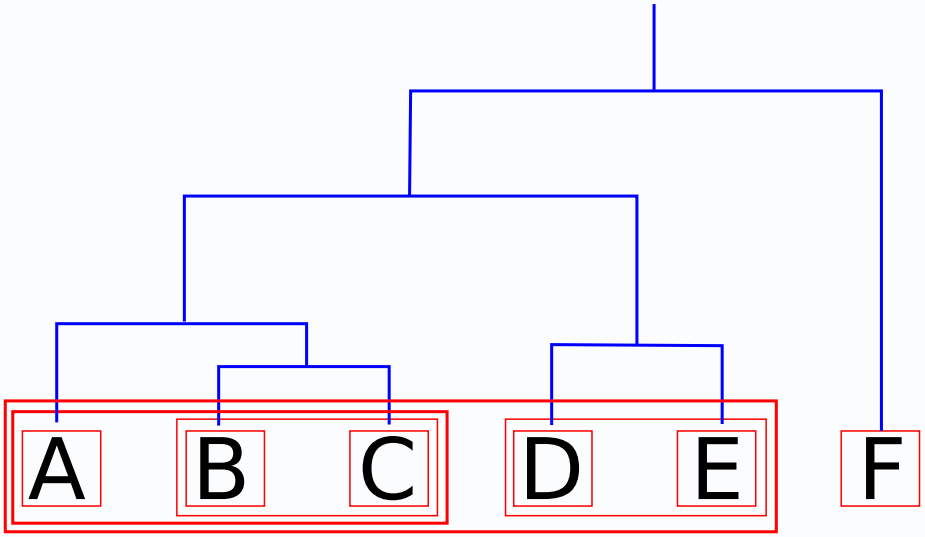

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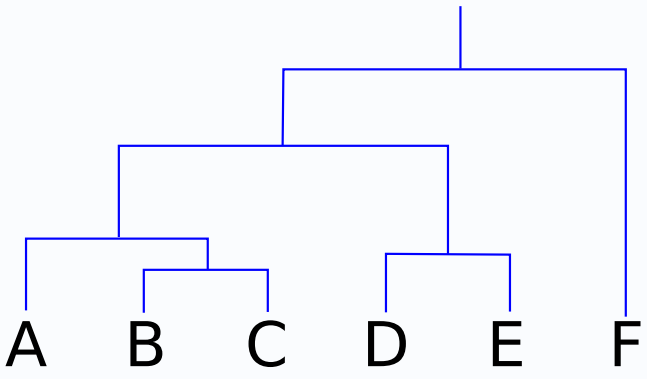
Understanding dendrograms

- provides a summary of the hierarchy
- a level in the tree (a.k.a. a node) corresponds to merging two classes
- the height of a level/node gives the "quality" of the merging



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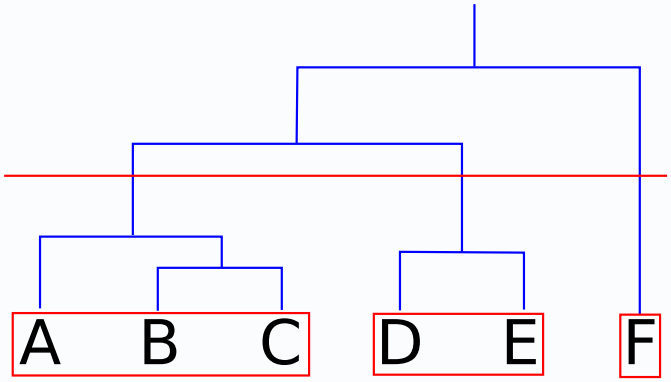

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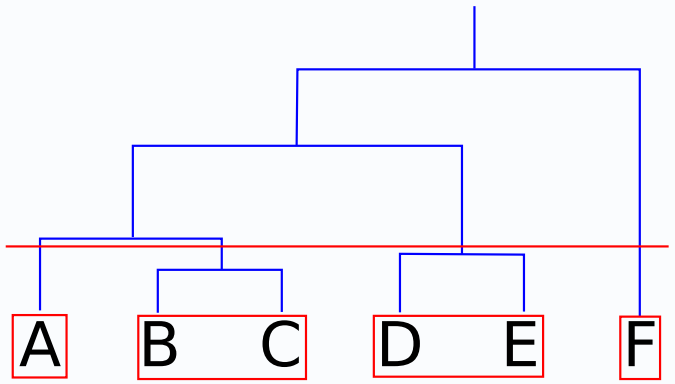

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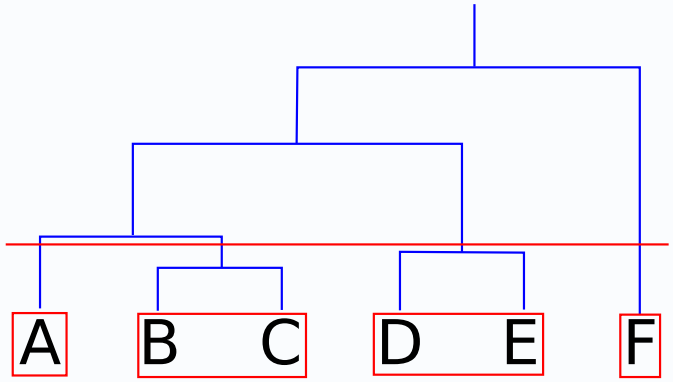

Common errors

- the dendrogram summarizes the **hierarchy** not the data
- do not interpret the order of the leaves: it is almost arbitrary (2^{N-2} orders for N leaves)

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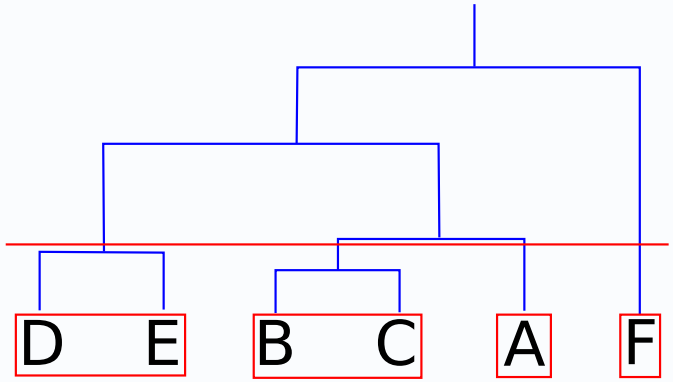

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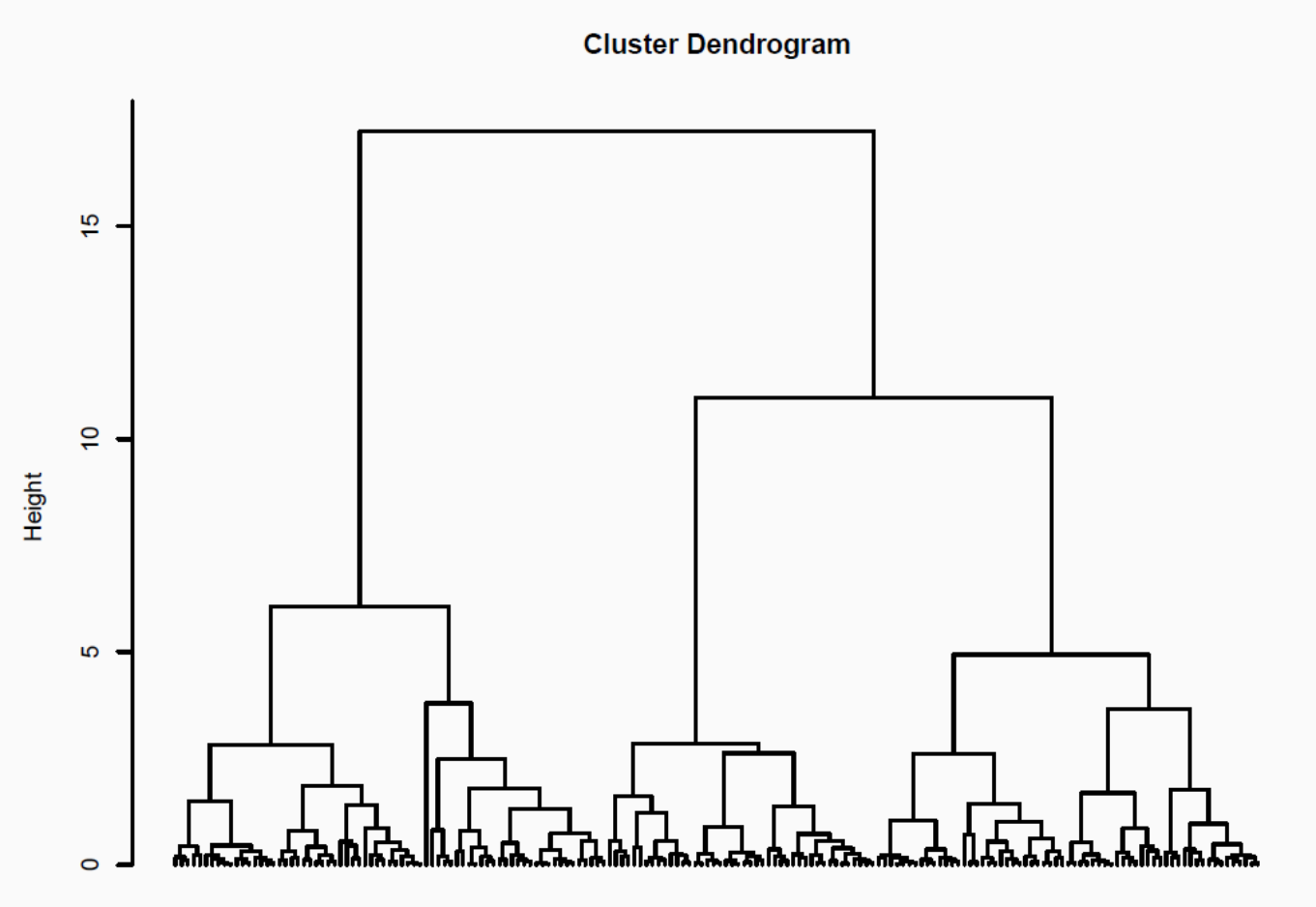
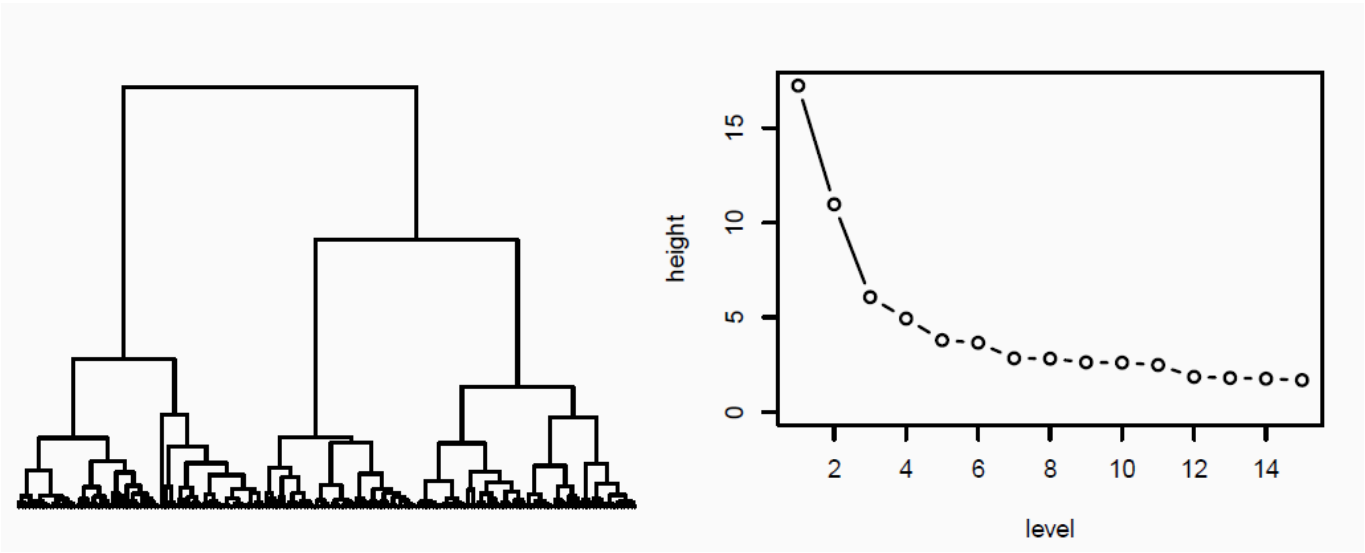
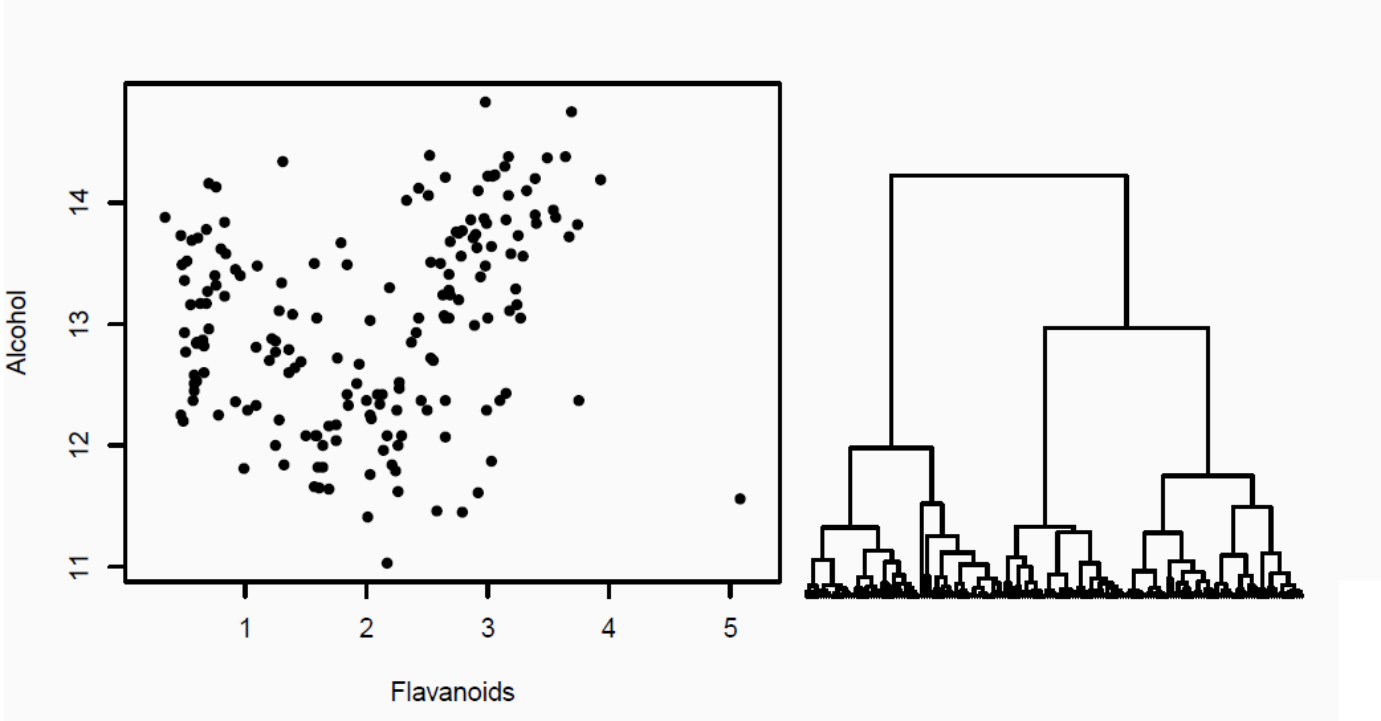


Hierarchical clustering



- look for "gaps" between levels: potential candidates for interesting partitions
- local partitions (i.e. branches) might also be interesting

Example



Hierarchical clustering

Agglomerative methods

- start with $P = \{\{1\}, \{2\}, \ldots, \{N\}\}$
- merge two classes in such a way that the resulting class is a *cluster*
- keep doing that until reaching $Q = \{\{1, \ldots, N\}\}$

Divisive methods

- start with $Q = \{\{1, \ldots, \{N\}\}$
- split one class into two sub-classes that are *clusters*
- keep doing that until reaching $P = \{\{1\}, \{2\}, \ldots, \{N\}\}$

Hierarchical clustering

Common elements

- those methods produce a hierarchy with N partitions
- they need a way to assess suitability of the classes as clusters
- dissimilarity based algorithms

Differences

- numerous agglomerative methods
- relatively few divisive ones
- very different computational problems:
  - agglomerative: $\Theta(N^2)$ potential merges at each step
  - divisive: $\Theta(2^{N-1})$ possible splits at the first step

Agglomerative methods

Core principle

- merge classes when they contain *similar* objects
- for singleton classes $\{\mathbf{X}_i\}$ and $\{\mathbf{X}_j\}$ use $d(\mathbf{X}_i, \mathbf{X}_j)$ to judge similarity
- key point: extend the dissimilarity to groups of objects

General algorithm

- initial partition: $\mathcal{P}^1 = \{\{1\}, \{2\}, \ldots, \{N\}\}$
- for k from 2 to N:
  - compute the dissimilarity between all current classes in $\mathcal{P}^{k-1}$
  - build $\mathcal{P}^k$ from $\mathcal{P}^{k-1}$ by merging the two least dissimilar classes

Inter-classes dissimilarities

Aggregation functions

Let A and B be two classes (of indices)

- single linkage (min)

$$d_S(A, B) = \min_{i \in A, j \in B} d(\mathbf{X}_i, \mathbf{X}_j)$$

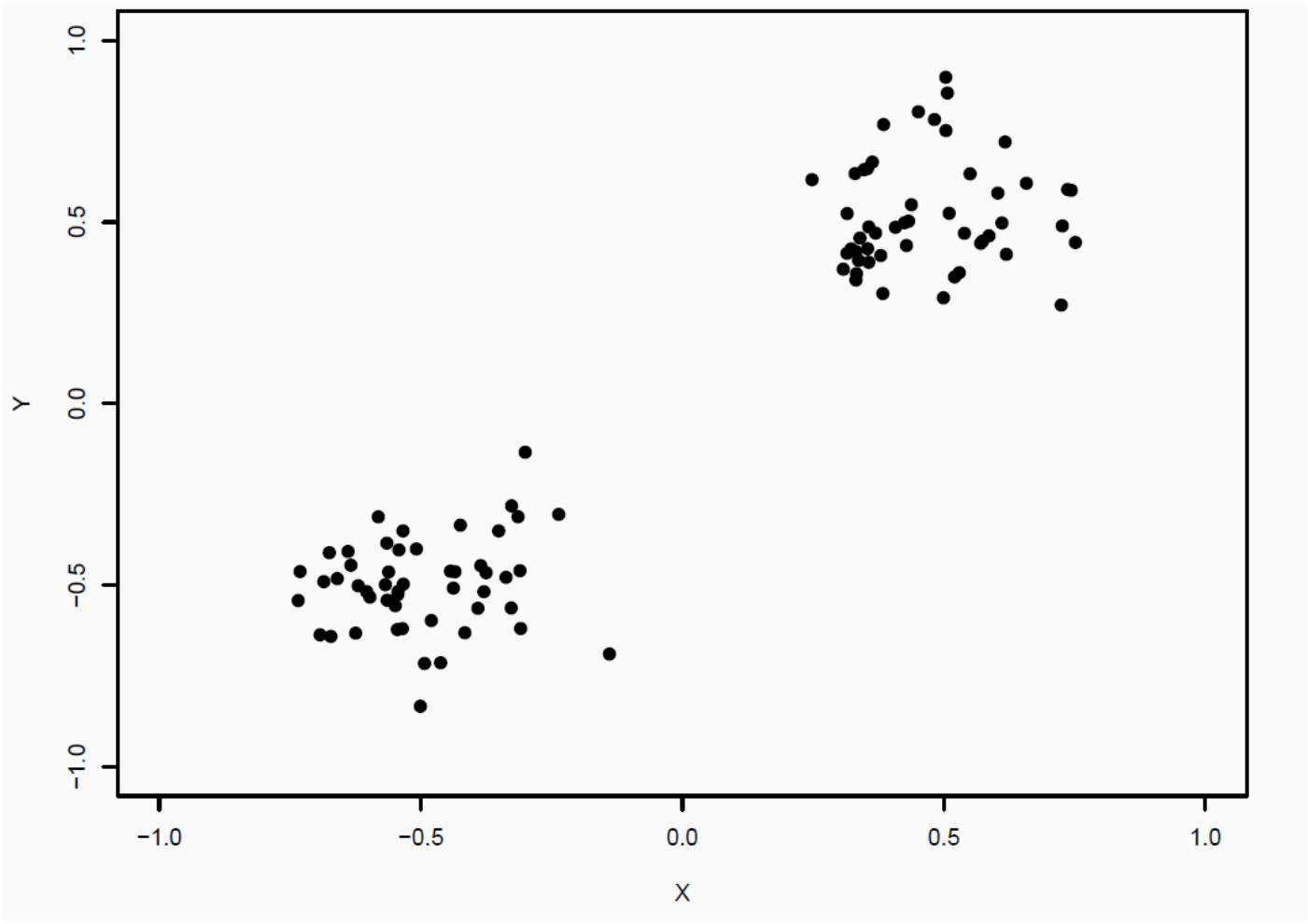
- complete linkage (max)

$$d_S(A, B) = \max_{i \in A, j \in B} d(\mathbf{X}_i, \mathbf{X}_j)$$

- average linkage

$$d_S(A, B) = \frac{1}{|A||B|} \sum_{i \in A, j \in B} d(\mathbf{X}_i, \mathbf{X}_j)$$

Illustration



Computational complexity

Naive approach

- $N-1$ steps
- step k searches for the best merge over $\frac{(N-k+1)(N-k)}{2}$ pairs of classes
- overall $\Theta(N^3)$ with adapted dissimilarity calculations
- standard implementation in many packages

Optimized solution

- simple priority list based solutions in $\Theta(N^2 \log N)$
- optimized algorithms in $\Theta(N^2)$ (see fastcluster in Python)
- notice that the dissimilarity must be calculated! For $(X) = \mathbb{R}^P$, this adds in general a $\Theta(N^2 P)$ cost.

In practice

Outline

1. choose a dissimilarity
2. choose an aggregation function
3. build the hierarchy
4. study the dendrogram:
   - the heights are the dissimilarities between merged classes
   - gaps can be seen as abrupt changes in merge qualities
   - visualization methods can be used to display the classes (e.g. Principal Component Analysis)

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1. choose a dissimilarity
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   - the heights are the dissimilarities between merged classes
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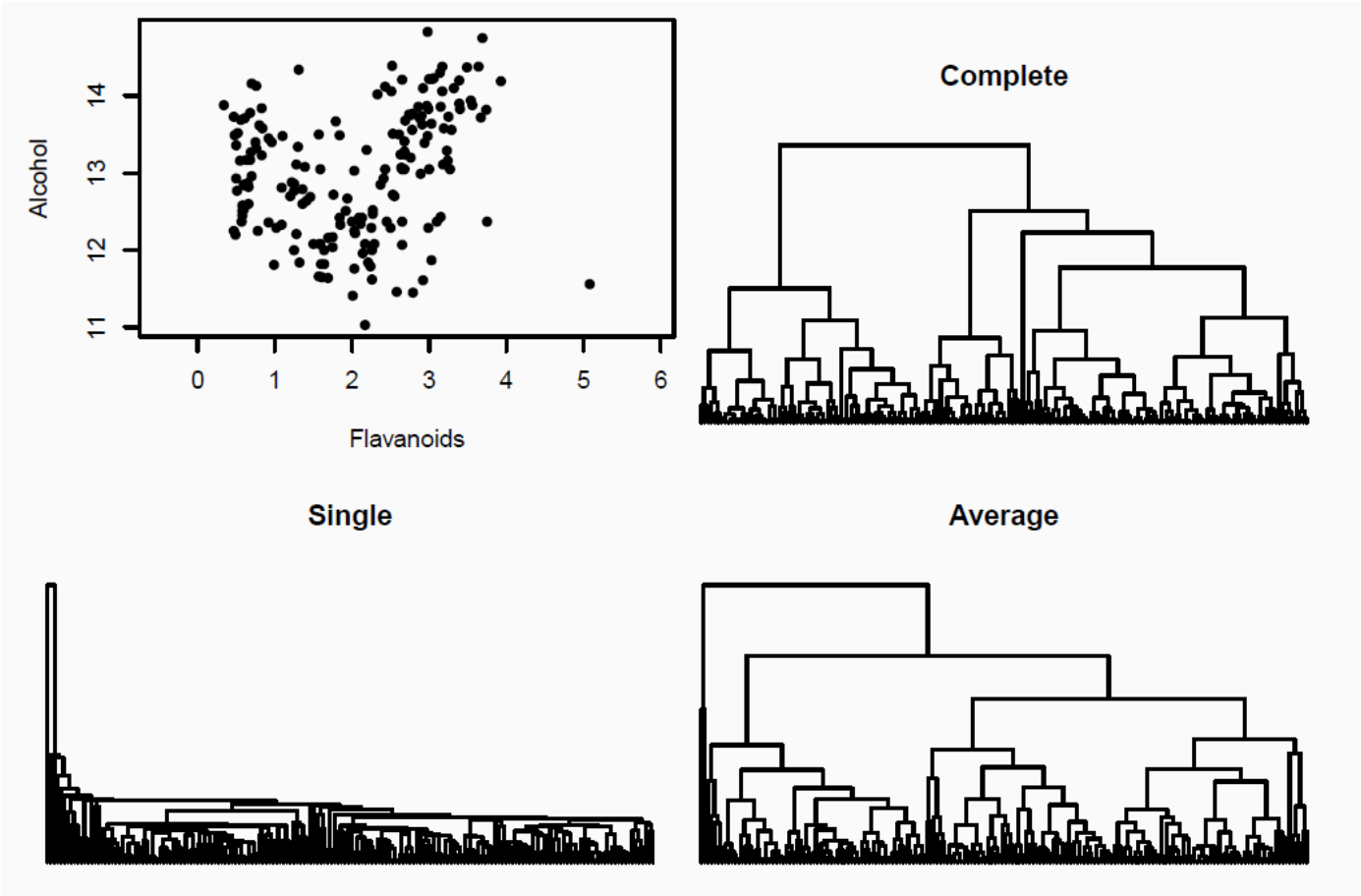
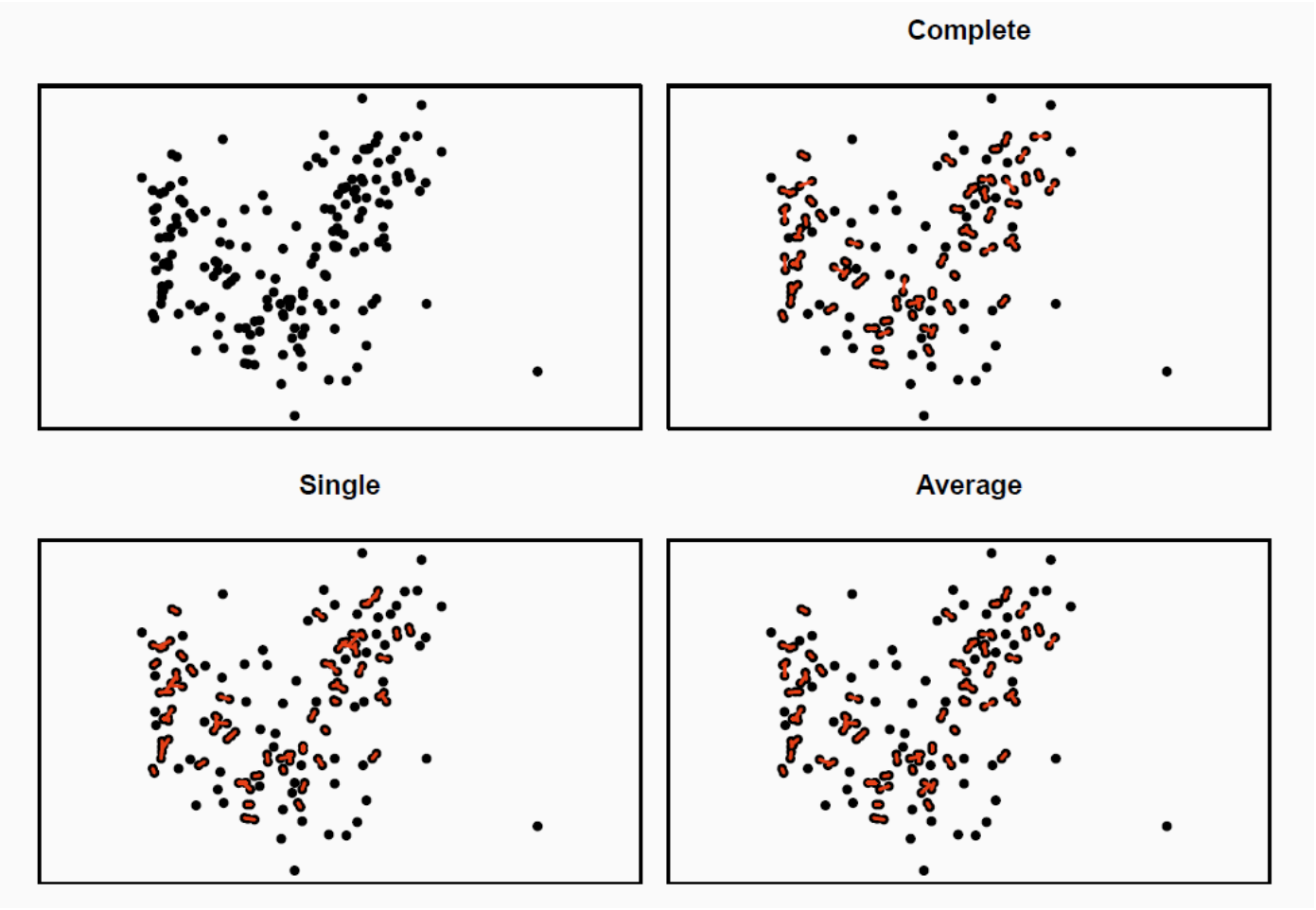
Limitations

- results strongly depend on both the dissimilarity and the aggregation function
- somewhat slow
- exploratory tool rather than clustering tool

Example


Alcohol
11
12
13
14
0
1
2
3
4
5
6
Flavanoids
Complete
Single
Average

Example



Example

Remarks

- initial behaviors are quite independent from the aggregation function
- single linkage:
  - tendency to produce "long" classes (chaining)
  - outlier isolation
- complete linkage:
  - balanced class sizes
  - classes can be close one to another (crowding)
- average linkage
  - average behavior...
  - resistant to noise but sensitive to dissimilarity transformation

Cluster quality

Classes as clusters

A partition provides a good *clustering* if its classes are

- homogeneous
- well separated

Criteria

- homogeneity via the diameter

$$D(A) = \max_{i \in A, j \in A} d((X)_i, (X)_j)$$

- separation via dissimilarity
- a good clustering: small diameters and large dissimilarities!

Aggregation function

Single linkage

- no control over $D(A)$ during the merge
- could end up with classes with very different diameters
- distances between classes can be very small relatively to their diameter

Complete linkage

- the diameter is the quality measure of a merge!
- however, no control at all over separability

Two extreme cases

with the average linkage in between...

In practice

General rules of thumb

- single linkage
  - gives frequently poor results
  - might be useful to spot outliers
- average linkage
  - gives generally good clusters
  - interesting compromise between diameter and separation
- complete linkage
  - maximally homogeneous classes
  - useful when there are no real clusters (i.e., clusters that can be easily separated)

Partition quality

Quality versus distance

- aggregation functions express the quality of a merge in terms of distances (min, max or average)
- alternative solution: express the quality using the resulting cluster

Within "variance"

- within variance of a class A (homogeneity measure)

$$W(A) = \frac{1}{|A|} \sum_{i \in A, j \in A} d((X)_i, (X)_j)$$

- total within variance of a partition $\mathcal{P} = \{A_1, \ldots, A_K\}$

$$W(\mathcal{P}) = \sum_{k=1}^{K} W(A_k)$$

Ward's method

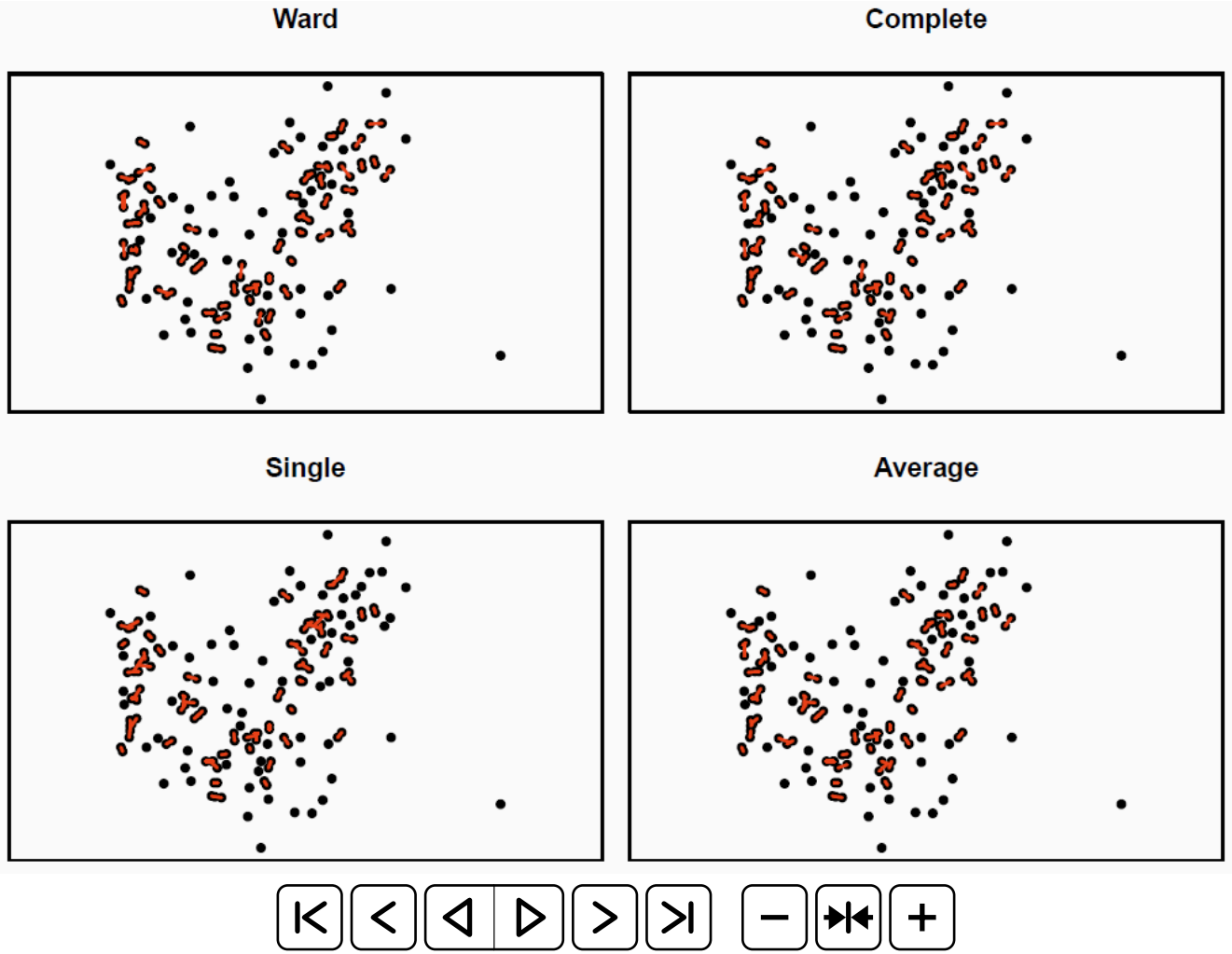
Optimizing the within variance

- same aggregative algorithm than with dissimilarities
- (non)quality of a merge:
  - the increase in within variance induced by merging A with B
  - local computation

$$\Delta_{(A,B)\to A\cup B} = W(A \cup B) - W(A) - W(B)$$

- can be seen as a greedy optimization of $W(\mathcal{P})$ at each stage of the hierarchy

Example



Discussion

Ward versus average

- somewhat related hierarchy
- Ward's method tend to favor more balanced class size
- Ward's method is closely related to other methods (e.g. K-means)
- outliers are more easily aggregated with other points in the Ward's method

Numerous other variants

- centroid and median linkage (distances between classes induced by prototype based representation)
- minimax linkage (enclosing ball diameter)
- etc.

Summary

Hierarchical clustering

- ✓ provide a hierarchy and a dendrogram
- ✓ applicable to any dissimilarity
- ✗ $\Theta(N^2)$ running time for aggregative methods
- ✗ not much guarantees about the final result
- ✗ inferior results for some methods (e.g. single linkage)

Generic questions

- dissimilarity?
- hierarchical method?
- cluster number?

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Criterion based clustering

Optimization point of view

- define a quality criterion for a clustering structure
- optimize this quality over "all" clustering structures
- typical example:
  - within variance as quality measure
  - optimized over partitions

Difficulties

- no obvious quality criterion for exploratory tasks
- very difficult discrete optimization problems (NP-hard in some cases)

Quantization point of view

Summarizing a data set

- key idea: represent a data set $\mathcal{D} = (\mathbf{X}_i)_{1 \leq i \leq N}$ by a smaller set of **prototypes** $\mathcal{D} = (\gamma_k)_{1 \leq k \leq K}$
- $\mathbf{X}_i$ is presented by a prototype: z_i is the index in $\{1, \ldots, K\}$ of this prototype
- natural risk associated to the problem

$$\varepsilon(\mathbf{\Gamma}, \mathbf{z}) = \sum_{i=1}^{N} d(\mathbf{X}_i, \gamma_{z_i}),$$

where $\mathbf{z} = (z_1, \ldots, z_N)$ and $\mathbf{\Gamma} = (\gamma_1, \ldots, \gamma_K)$.

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  where $\mathbf{z} = (z_1, \ldots, z_N)$ and $\mathbf{\Gamma} = (\gamma_1, \ldots, \gamma_K)$.
- natural clustering interpretation: $\mathbf{z}$ defines a partition of $\{1, \ldots, N\}$ into K classes!

Optimization

Difficult!

- minimizing $\varepsilon(\mathbf{\Gamma}, \mathbf{z})$ is a combinatorial problem
- for standard d (such as the squared Euclidean distance when $\mathcal{X} = \mathbb{R}^P$) this is NP-hard

Simple sub-problems

- minimizing $\varepsilon(\mathbf{\Gamma}, \mathbf{z})$ with respect to $\mathbf{z}$ is easy

$$\arg\min_{\mathbf{z}} \varepsilon(\mathbf{\Gamma}, \mathbf{z}) = \left(\arg\min_{k \in \{1,\dots,K\}} d(\mathbf{X}_1, \gamma_k), \dots, \arg\min_{k \in \{1,\dots,K\}} d(\mathbf{X}_N, \gamma_k) \right)$$

- if prototypes are restricted to be elements of the data set $\mathcal{D}$, optimizing with respect to $\mathbf{\Gamma}$ is also easy

$$\arg\min_{\mathbf{\Gamma}} \varepsilon(\mathbf{\Gamma}, \mathbf{z}) = \left(\arg\min_{\gamma_k \in \mathcal{D}} \sum_{i, z_i = k} d(\mathbf{X}_i, \gamma_k) \right)_{1 \leq k \leq K}$$

Coordinate descent

A.k.a. alternating optimization

a simple minimization algorithm for a function $F(u, v)$:

select u_0 randomly
$k \leftarrow 1$
repeat
  $v_k = \arg\min_v F(u_{k-1}, v)$
  $u_k = \arg\min_u F(u, v_k)$
  $k \leftarrow k + 1$
until convergence

Properties

- converges to a local minimum
- but not to a global one
- improved by multiple restarts

K-medoids

Algorithm

select $\mathbf{\Gamma}$ as a random subset of $\mathcal{D}$

repeat

$z_i \leftarrow \arg\min_{k \in \{1,\dots,K\}} d(\mathbf{X}_i, \gamma_k)$ ▷ assignment phase

$\gamma_k \leftarrow \arg\min_{\gamma \in \mathcal{D}} \sum_{i, z_i = k} d(\mathbf{X}_i, \gamma)$ ▷ representation phase

until convergence

Comments

- one of the most well known clustering algorithm for arbitrary dissimilarities
- complexity $\Theta(NK) + \Theta(N^2)$ (assuming d is known)
- complex quantization effects when N is small ("data holes")

Euclidean/Hilbert case

When $\mathcal{X} = \mathbb{R}^P$

- one uses in general $d(\mathbf{X}_i, \mathbf{X}_j) = \|\mathbf{X}_i - \mathbf{X}_j\|^2$
- then we do need to restrict γ to elements of $\mathcal{D}$
  - we have
$$\varepsilon(\mathbf{\Gamma}, \mathbf{z}) = \sum_{k=1}^{K} \sum_{i, z_i = k} \|\mathbf{X}_i - \gamma_k\|^2$$
  - and thus
$$\arg\min_{\mathbf{\Gamma}} \varepsilon(\mathbf{\Gamma}, \mathbf{z}) = \left(\frac{1}{s_k} \sum_{i, z_i = k} \mathbf{X}_i \right)_{1 \leq k \leq K},$$
    where $s_k = |\{i | z_i = k\}|$
- identical results when $\mathcal{X}$ is a Hilbert space (a RHKS for instance)

K-means algorithm

Algorithm

select $\mathbf{\Gamma}$ as a random subset of $\mathcal{D}$

repeat

$z_i \leftarrow \arg\min_{k \in \{1,\dots,K\}} \|\mathbf{X}_i - \gamma_k\|^2$ ▷ assignment phase

$\gamma_k \leftarrow \frac{1}{s_k} \sum_{i, z_i = k} \mathbf{X}_i$ ▷ representation phase

until convergence

Comments

- one of the most well known clustering algorithm
- complexity $\Theta(NKP)$
- numerous variants and improvements (kmeans++ for instance)

Initialization

Standard solution

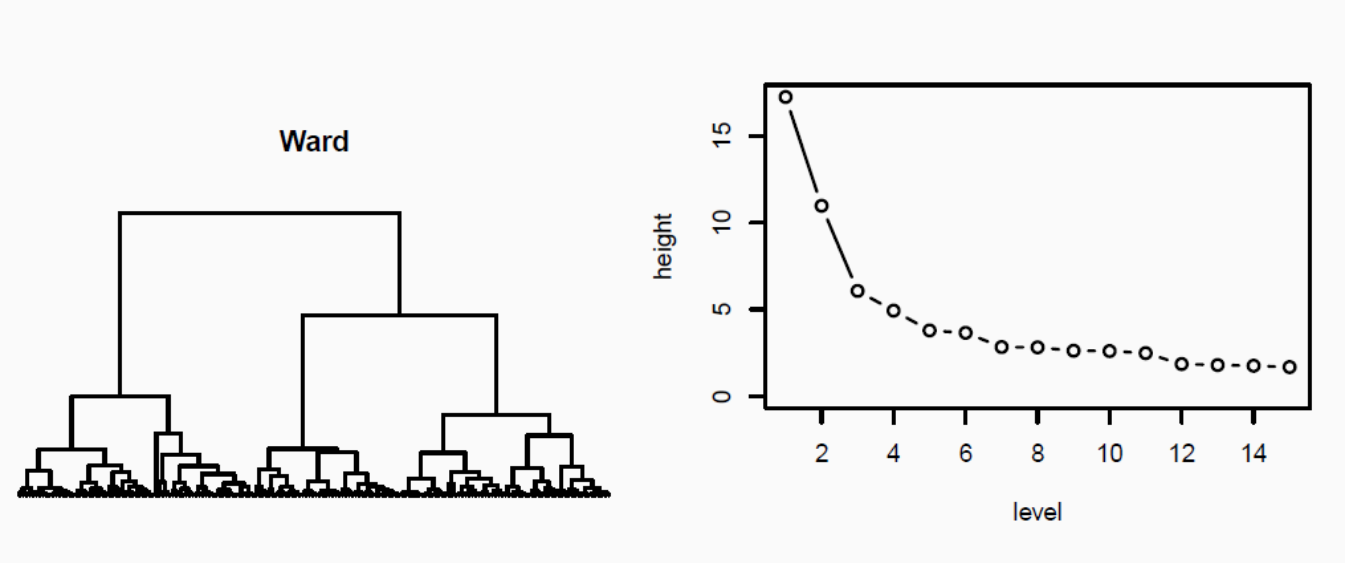
- random prototypes chosen uniformly at random in the data set without replacement
- random clusters do not work properly

k-means++ (Arthur & Vassilvitskii, 2007)

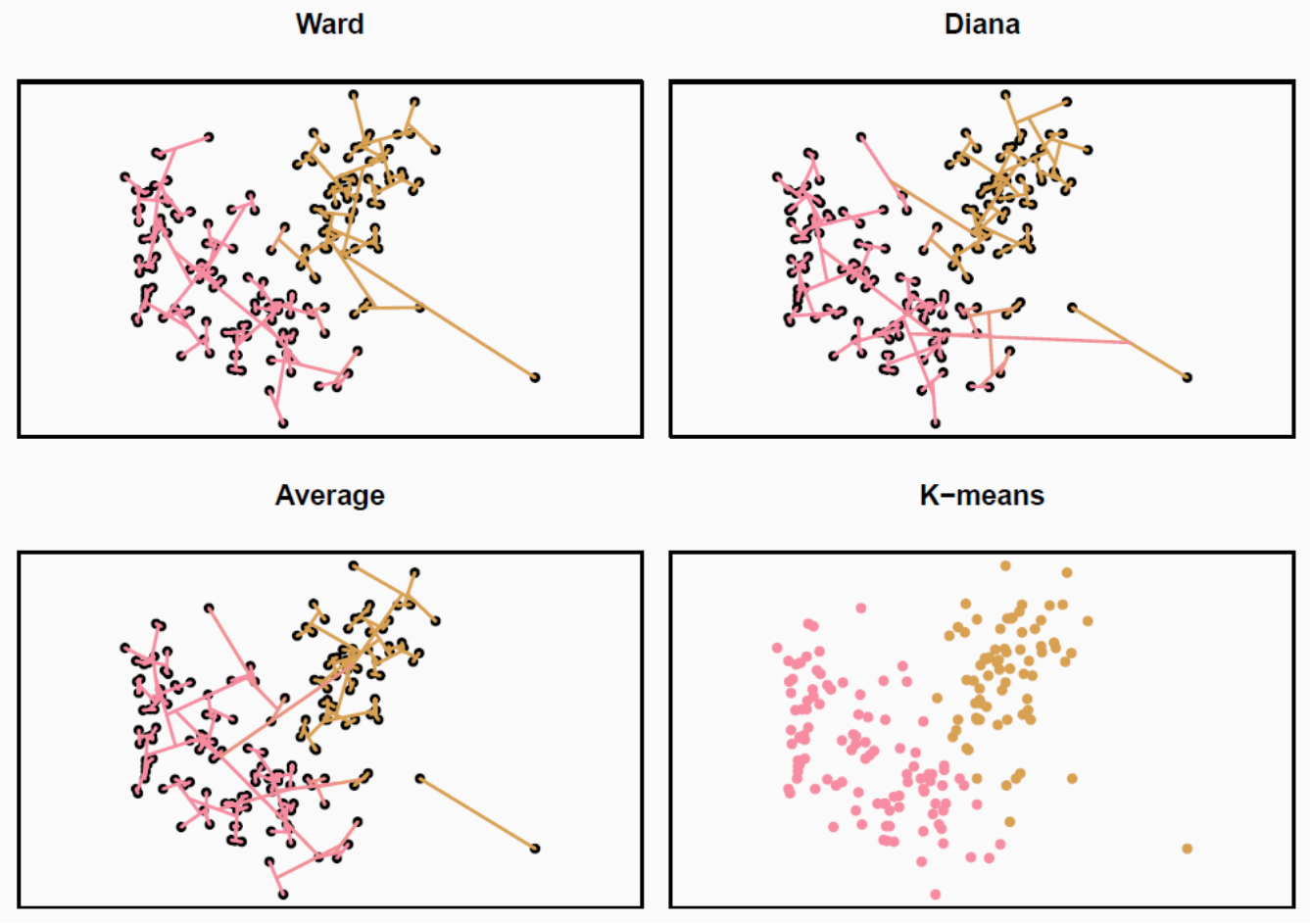
1. γ_1 chosen uniformly at random in the data set
2. for k in 2 to K:
   1. for $\mathbf{X}_i \in \mathcal{D} \setminus \{\gamma_1, \dots, \gamma_{k-1}\}$ compute $p_i = \frac{\min_{k \in \{1,\dots,k-1\}} d(\mathbf{X}_i, \gamma_k)^2}{\sum_{j \neq i} \min_{k \in \{1,\dots,k-1\}} d(\mathbf{X}_j, \gamma_k)^2}$
   2. chose γ_k in $\mathcal{D} \setminus \{\gamma_1, \dots, \gamma_{k-1}\}$ according to the probabilities $(p_i)_i$

- theoretical guarantees
- practical efficiency

Example



Example



K-means and general clustering

Variance decomposition

- for any $C \subset \{1, \dots, N\}$

$$\sum_{i \in C} \sum_{j \in C} \|\mathbf{x}_i - \mathbf{x}_j\|^2 = 2|C| \sum_{i \in C} \left\| \mathbf{x}_i - \widehat{\mathbf{x}}_C \right\|^2,$$

where

$$\widehat{\mathbf{x}}_C = \frac{1}{|C|} \sum_{i \in C} \mathbf{x}_i$$

- and therefore if $C_k = \{i \in \{1, \dots, N\} \mid z_i = k\}$

$$\min_{\boldsymbol{\Gamma}} \sum_{i=1}^{N} \|\mathbf{x}_i - \gamma_{z_i}\|^2 = \frac{1}{2} \sum_{k=1}^{K} \frac{1}{|C_k|} \sum_{i \in C_k} \sum_{j \in C_k} \|\mathbf{x}_i - \mathbf{x}_j\|^2$$
$$= \frac{1}{2} W(\{C_1, \dots, C_k\})$$

Variance and clustering

Quantization and within variance

- minimizing the quantization error is equivalent to maximizing the total within variance
- K-means is very close to Ward's method!

Total variance

- the total variance is $\sum_{i=1}^{N} \left\| \mathbf{x}_i - \widehat{\mathbf{x}}_{\mathcal{D}} \right\|^2$
- we have

$$\sum_{i=1}^{N} \left\| \mathbf{x}_i - \widehat{\mathbf{x}}_{\mathcal{D}} \right\|^2 = \min_{\Gamma} \sum_{i=1}^{N} \left\| \mathbf{x}_i - \widehat{\mathbf{x}}_{C_{\gamma_{z_i}}} \right\|^2 + \sum_{k=1}^{K} |C_k| \left\| \widehat{\mathbf{x}}_k - \widehat{\mathbf{x}}_{\mathcal{D}} \right\|^2$$

$$= \frac{1}{2} W(\{C_1, \ldots, C_k\}) + \frac{1}{2} \sum_{k=1}^{K} \sum_{k'=1}^{K} |C_k||C_{k'}| \left\| \widehat{\mathbf{x}}_k - \widehat{\mathbf{x}}_{k'} \right\|^2$$

Variance and clustering

Between variance

- $B(C_1, \ldots, C_k) = \frac{1}{2} \sum_{k=1}^{K} \sum_{k'=1}^{K} |C_k||C_{k'}| \left\| \widehat{\mathbf{X}}_k - \widehat{\mathbf{X}}_{k'} \right\|^2$
- weighted pairwise distances between prototypes
- measures how spread the clusters are

Within and between variance

- total variance = within variance + between variance
- the total variance does not depend on the clustering
- by minimizing the within variance, one maximizes the between variance!
- clusters are both compact and well separated (at least at the prototype level)

Summary

K-means

- ✓ clear quantification interpretation
- ✓ compact clusters with separated prototypes
- ✓ $\Theta(NKP)$ running time
- ✓ efficient initialization strategy (k-means++)
- ✗ K?
- ✗ spherical clusters which might be very close one to another